University of Idaho
College of Engineering

CS 404/504, ME 404/504 PLC PROGRAMMING AND AUTOMATION FALL 2022

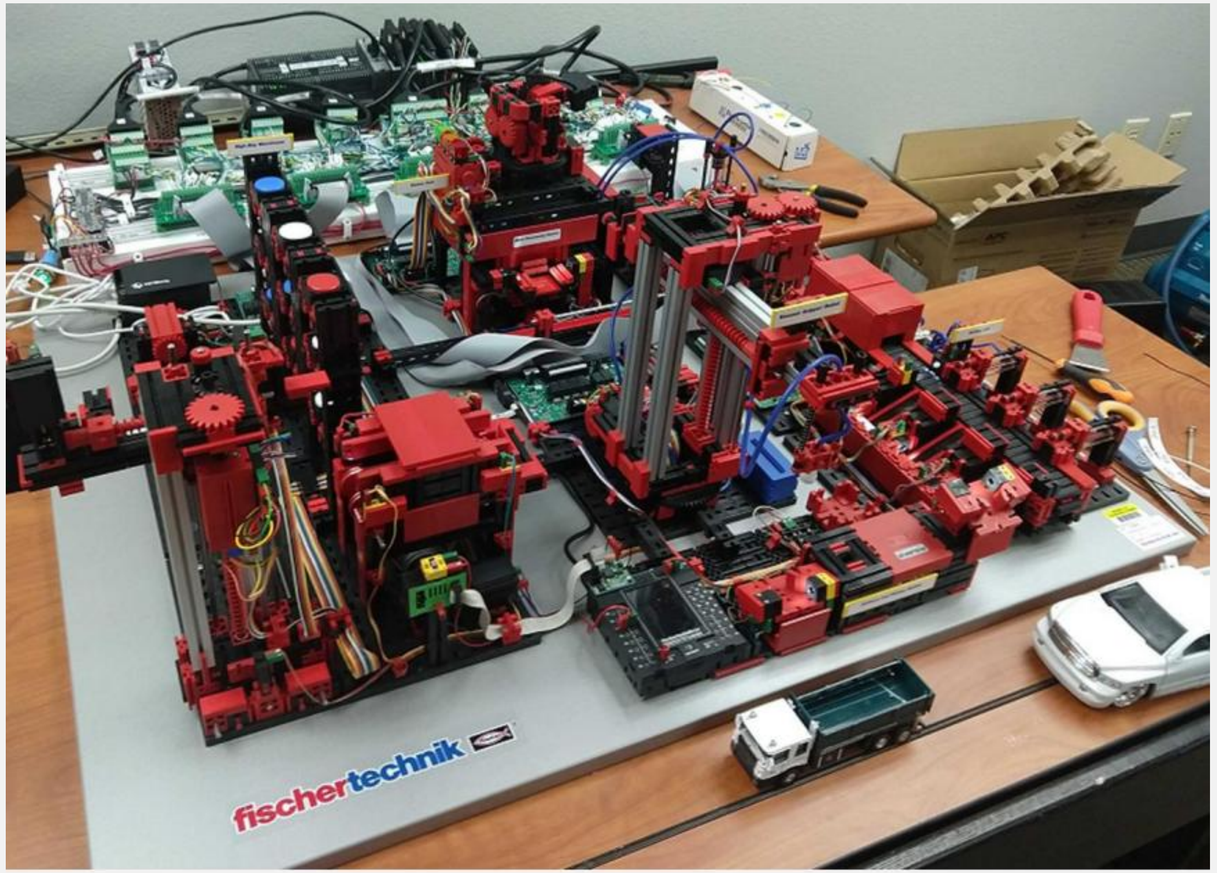
LECTURE 18 – MINI FACTORY OVERVIEW

INSTRUCTOR: DR JOHN SHOVIC
LECTURER: NIKOLAI TIONG
TA: JACOB FRIEDBERG

FISCHERTECHNIK MINI FACTORY

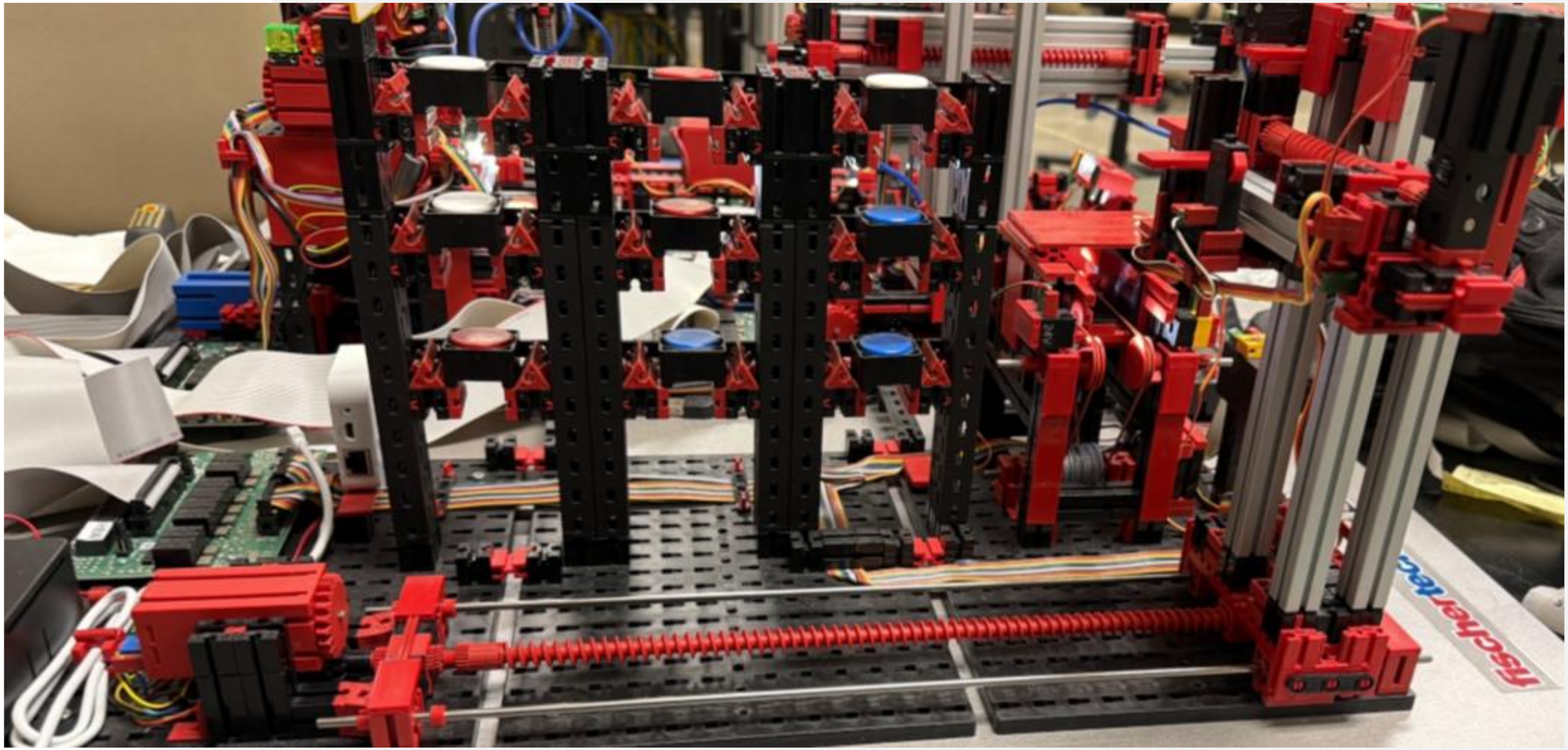
THOSE THINGS ON THE TABLES YOU SEE EVERY CLASS

- Consists of three main parts
  - Mini Factory itself
  - BRX PLC
  - Maple Systems HMI
  - All the wiring to connect it together – PCB
    - Ribbon, ZIPLink cables, network



MINI FACTORY MODULES

HIGH BAY WAREHOUSE



MINI FACTORY MODULES

HIGH BAY WAREHOUSE (HBW) I/O

- Inputs
  - Reference switches – reached a physical limit with movement, used to reset counters
  - Ligh Barriers – Detects when the beam breaks
  - Encoders – detect movement of motors
- Outputs
  - Motors – move the crane around and the conveyor belt
  - PWM Control for Motors – vary speed

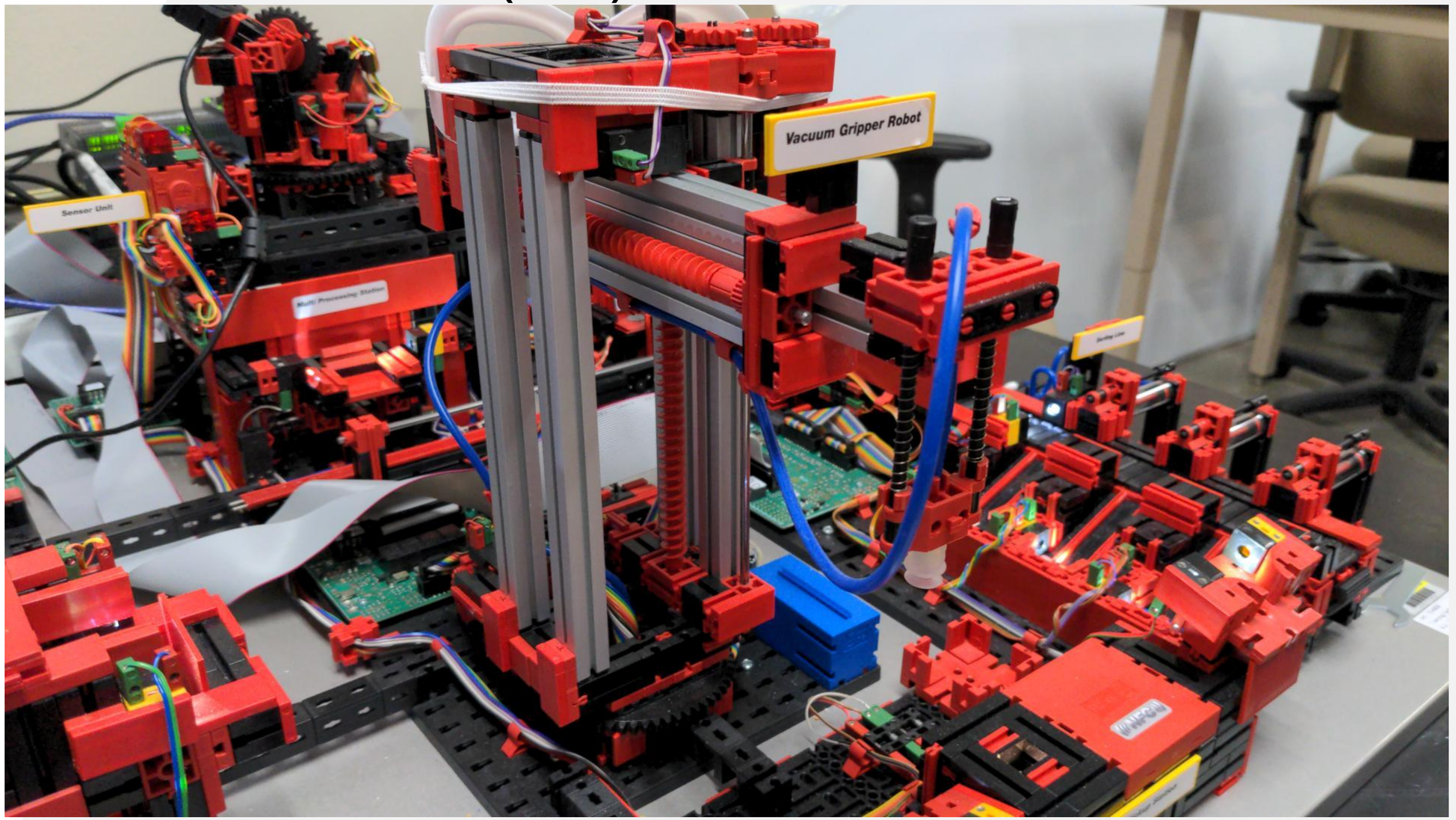
MINI FACTORY MODULES

HIGH BAY WAREHOUSE (HBW) HAZARDS

- Moving the forklift with cantilever extended can cause it to collide with the side of the warehouse, bending it, or knocking pucks off the racks.
- If you are not lined up properly for each position you can hit the the side of the slot pushing the warehouse over
- If you do not have a signal from the VGR that the crane is clear, you can roll back a empty pallet and hit the VGR as its trying to lift away

MINI FACTORY MODULES

VACUUM GRIPPER ROBOT (VGR)



MINI FACTORY MODULES

VACUUM GRIPPER ROBOT (VGR) I/O

- Inputs
  - Reference switches – reached a physical limit with movement, used to reset counters
  - Encoders – detect movement of motors
- Outputs
  - Motors – move the crane around and the conveyor belt
  - PWM Control for Motors – vary speed
  - Compressor and Vacuum – picks up disks

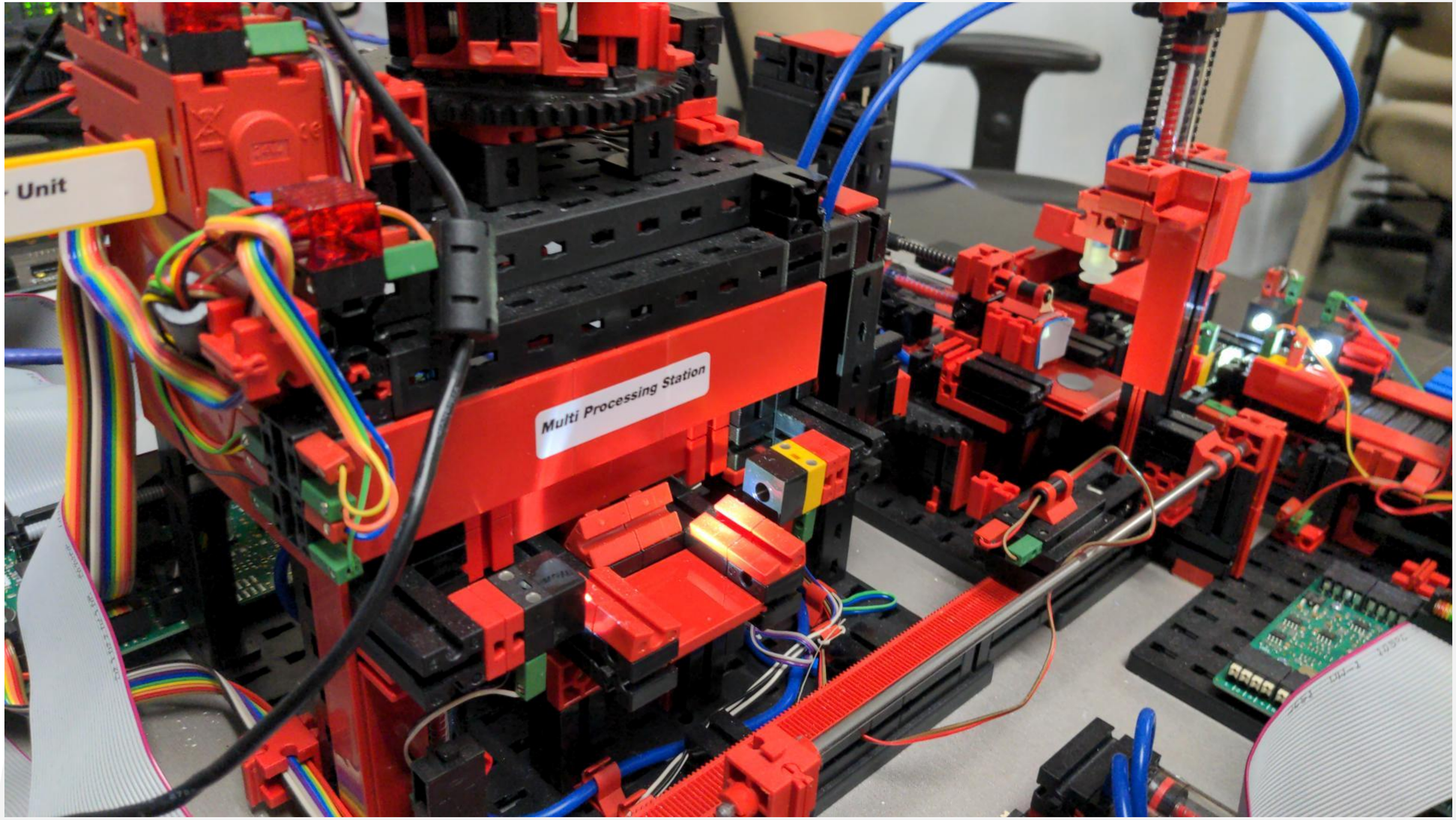
MINI FACTORY MODULES

VACUUM GRIPPER ROBOT (VGR) HAZARDS

- Rotating the crane before retracting the arm will cause it to collide with the SSC camera or the HBW rack
- Rotating the crane with the crane lowered can cause it to collide with objects close to the ground
- Extending the crane when there isn’t enough space will cause flexing
- Lowering the vacuum gripper in the wrong place can cause it to get stuck – the entire VGR crane can be lifted off its mounting and rotated to fix this
- The vacuum gripper may not pick up a disk because there wasn’t one in the pallet from the HBW in the first place

MINI FACTORY MODULES

MULTI-PROCESSING STATION AND OVEN (MPO)



MINI FACTORY MODULES

MULTI-PROCESSING STATION AND OVEN (MPO) I/O

- Inputs
  - Reference switches – reached a physical limit with movement or a specific position
  - Light barrier – detect disk got dropped onto the oven feeder
- Outputs
  - Motors – move the vacuum gripper, conveyor belt and oven feeder
  - Motor PWM – control speed of vacuum gripper motor
  - Compressor and Vacuum – picks up disks, lifts up and retracts oven door, push disk onto conveyor belt
  - Light – Indicate oven is on

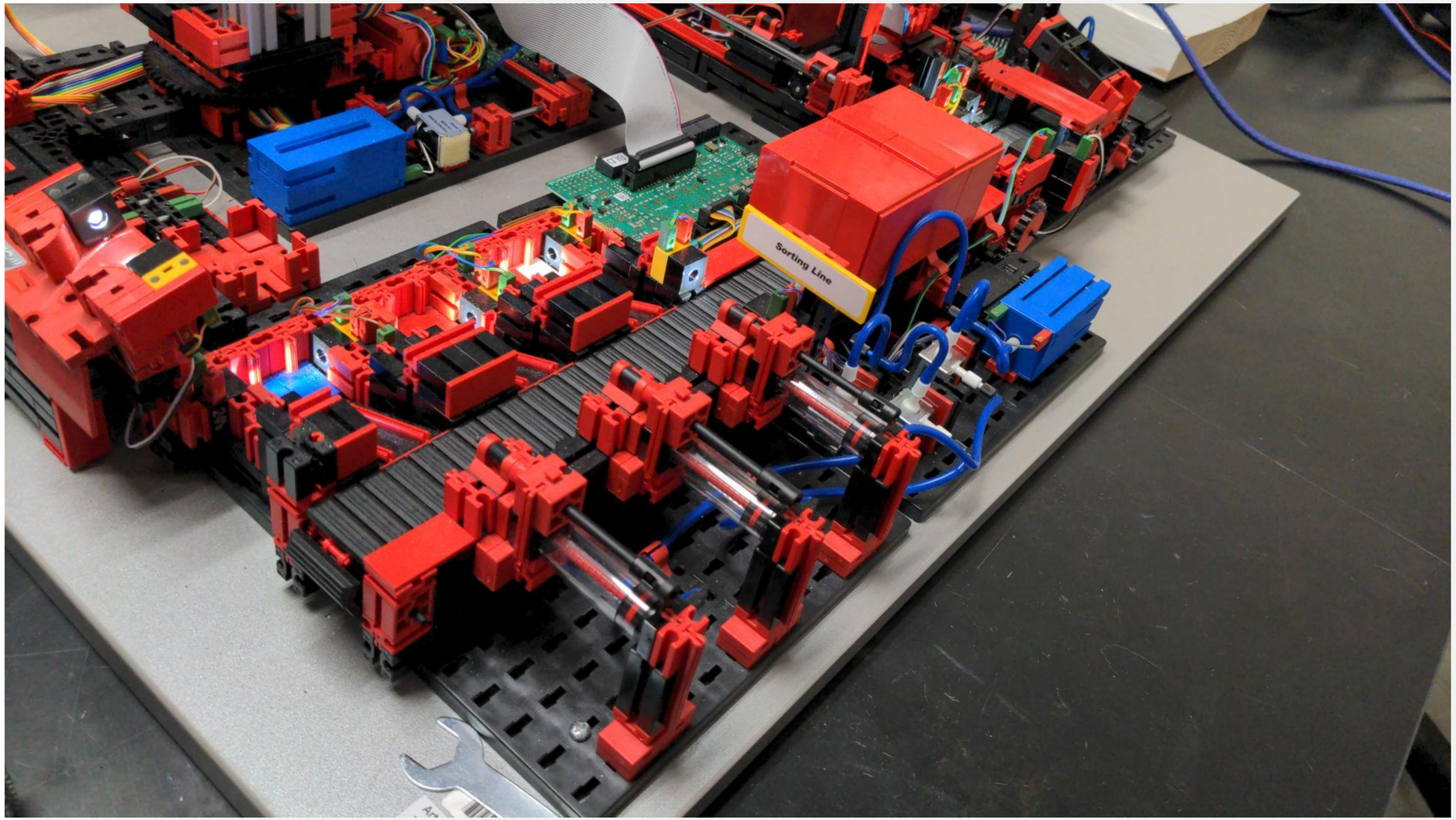
MINI FACTORY MODULES

MULTI-PROCESSING STATION AND OVEN (MPO) HAZARDS

- The oven feeder can be retracted or extended with the door lowered
  - This will cause a lot of flexing on the door
  - Turn the PLC off as soon as you notice this happening
  - Make sure the door is raised before you retract or extend the feeder
- The disk can get knocked off the turntable when it rotates, or get stuck rotating
  - Must have a way to detect if this happens
- The disk may not make it onto the conveyor belt when the valve pushes it
  - Must detect this, possibly same way as the disk falling off

MINI FACTORY MODULES

SORTING LINE AND COLOR DETECTION (SLD)



MINI FACTORY MODULES

SORTING LINE AND COLOR DETECTION (SLD) I/O

- Inputs
  - Light barrier – detect disks at several points
  - Color sensor – analog sensor that detects what color the object is
- Outputs
  - Motors –conveyor belt
  - Compressor and Vacuum – pushes disks into the respective bays

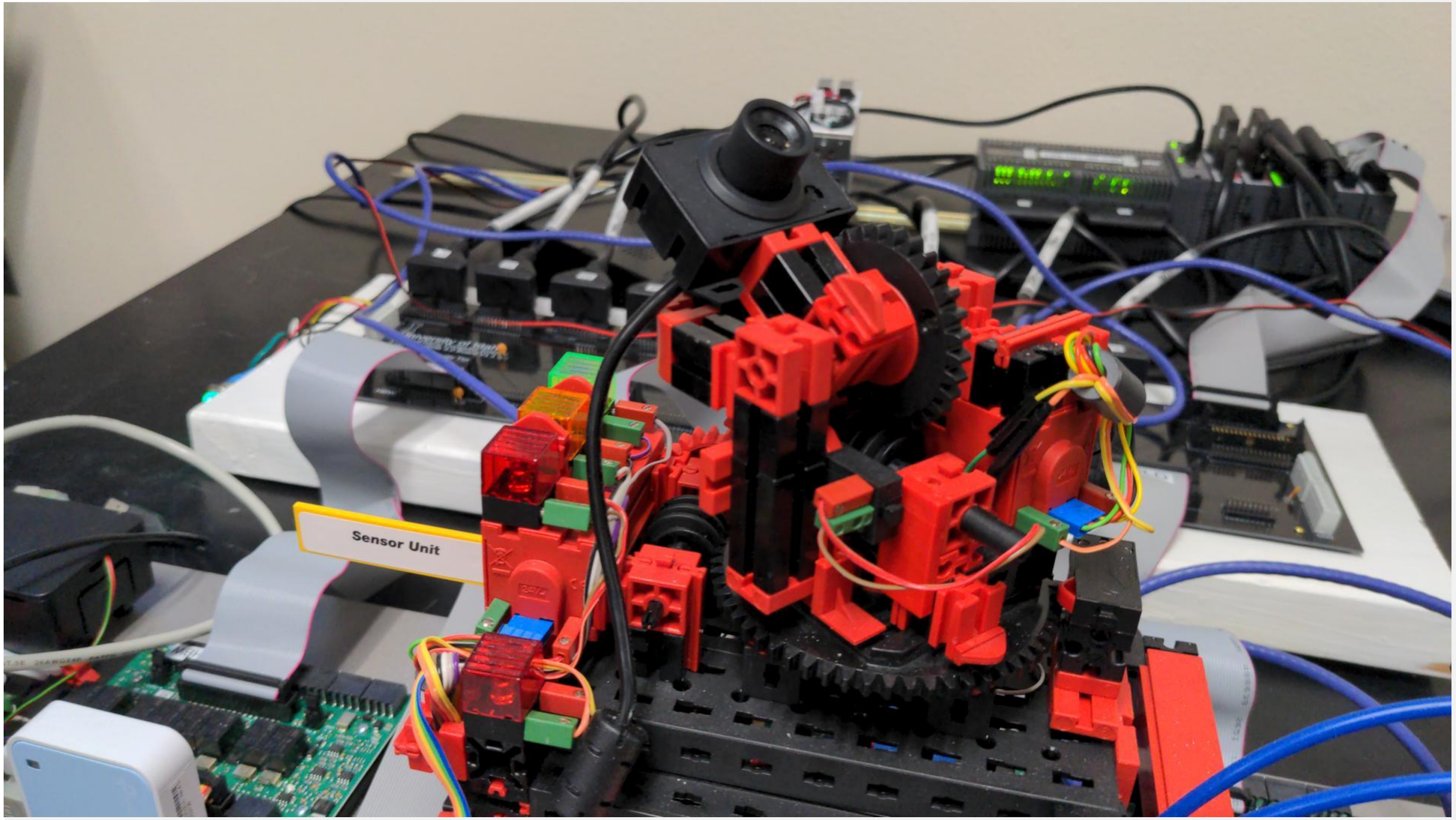
MINI FACTORY MODULES

SORTING LINE AND COLOR DETECTION (SLD) HAZARDS

- The transition between the MPO conveyor belt and the SLD one may not transfer the disk over
- The conveyor belt can keep moving forever
- The valves don’t push the disk in because it wasn’t detecting the color of the disk
  - Disk just rolls off the end of the belt
- The SLD module can start automatically if you put a disk or other obstruction in front of the first light barrier

MINI FACTORY MODULES

SENSOR SYSTEM AND CAMERA (SSC)



MINI FACTORY MODULES

SENSOR SYSTEM AND CAMERA (SSC) I/O

- Inputs
  - Reference switches – reached a physical limit with camera mount
  - Light barrier – detect disks in several spots
  - Color sensor – analog sensor that detects what color the object is
  - Encoder – measure and detect movement of camera mount
- Outputs
  - Motors – move the camera mount
  - LEDs – use for displaying status of entire system

MINI FACTORY MODULES

SENSOR SYSTEM AND CAMERA (SSC) HAZARDS

- The limit switches on rotating the camera don’t physically block movement
  - You can rotate the camera around indefinitely if you ignore the limit switches, ask Jacob what happens if you do that
- LEDs need to be explicitly turned off on reset, otherwise they will remember their last status

MINI FACTORY GOAL

MINI FACTORY PRODUCTION CYCLE

- A complete cycle of the Mini factory is as follows.
  - The HBW crane goes to a position in the warehouse grid, lines itself up then extends its arm below a pallet of material, lifts the material slightly, then retracts to gain control of the pallet. The crane then moves toward the transfer belt close to where the crane started. The crane lowers, and extends the pallet so that it sits on the belt. The belt is activated and the pallet appears on the other side for the VGR to grab. When the puck has been grabbed and the VGR is clear, the pallet will move back toward the crane and needs to be picked up and placed back in the same spot it was taken from.
  - The VGR crane waits for a pallet to show up at the transfer chain. When it does, the VGR will rotate itself counterclockwise toward the transfer chain, extend the crane, lower the crane to above the puck turn on the compressor, and then lower itself into the puck on the pallet. Once the silicone tip has pressed itself into the puck to create a seal the valve for the VGR will be enabled and the gripper will lift up, rotate clockwise, and lower itself onto the awaiting MPO feeder tray, and drop the puck. It will then move out of the way.

MINI FACTORY GOAL

MINI FACTORY PRODUCTION CYCLE

- The MPO will open the oven door, pull the feeder inside, close the door, and turn on the oven light and wait some amount of time to cook. Once cooking is done the door is opened, the feeder is extended, and the vacuum gripper robot is sent horizontally to go over the feeder, turns on its compressor, lowers the silicone tip into the puck, turns on the valve to create suction and then lifts itself to begin carrying the puck. The puck is moved to the turn table where it is dropped off by the gripper robot. The turn table turns to the saw cut position, and cuts the material for some amount of time. Then the turn table turns to the sorting line and pushes the puck onto it via a compressor and valve. The puck is then allowed to move on a conveyor belt before hitting a dip in the belt and transferring to the sorting line (SLD). You will want to keep the belt running until the puck is fully off the belt. It can get stuck in the dip.
- The SLD runs its conveyor, measures the light value of the puck as it goes through the color reading chamber, then based on timing pushes the puck into its respective color sorted bin.
- This cycle repeats for all pucks in the HBW. The HMI must have controls to perform this cycle for a single selected puck, and for all pucks in a hands free manner.